# Mohamad Deifallah

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## Work Experience

**Junior Information Management Technologist Intern** Oct 2024 – Dec 2024
*Natural Resources Canada* *Ottawa, ON*
- Designed a dynamic sub-report selection table in Microsoft Access, eliminating 3+ manual steps per report cycle
- Developed 5+ VBA automation scripts to streamline database workflows and reduce repetitive configuration tasks
- Implemented SQL-based filtering and redesigned 3 core forms in collaboration with project leads, cutting query turnaround time by 15%

**Software Developer Intern** Jan 2024 – Apr 2024
*National Research Council Canada* *Ottawa, ON*
- Built a responsive web application in HTML, CSS, and JavaScript used across a 4-month active research initiative
- Authored 3 research proposals covering strategic planning, compliance review, and feasibility analysis to support active NRC initiatives

**Front-End Developer, React.js** May 2023 – Aug 2023
*LineupQ* *Remote/ New York, NY*
- Delivered full-stack React and Node.js features across weekly Agile sprints in a production codebase serving real users
- Integrated Stripe payment authentication supporting secure, PCI-compliant transactions for a live product
- Improved frontend performance by 65% through Lighthouse-optimized design and UX enhancements

## Project Experience

**EchoRoam** | *C, QNX RTOS, HTML, Javascript, Docker, CSS* Mar 2025
- Built an autonomous rescue vehicle prototype on Raspberry Pi 4 using QNX RTOS and ultrasonic sensors for real-time obstacle detection and route mapping

**D.U.C.K.I. Research Project** | *Logistic Regression, Linear Regression, Neural Networks* Jan 2025
- Proposed and presented an AI-driven adaptive learning system featuring voice input, mouth movement analysis, and interactive responses

**E-Commerce Application** | *Java, IntelliJ IDEA* Nov 2024
- Built a virtual storefront with object-oriented design patterns, a modular architecture, and a text-based checkout interface

**Photo Editing Program** | *Python, Wing 101* Apr 2024
- Created an interactive photo-editing tool with pixel manipulation, digital color processing, and a modular text-based interface

## Education

**Carleton University** Apr 2026
*Bachelor of Computer Science Major* *Ottawa, ON*
- **Relevant Coursework:** Data Structures, Algorithms, Databases, Design of User Interfaces, Software Requirements Analysis, Software Design and Architecture, Intro to Artificial Intelligence

## Leadership Experience

**Faculty of Engineering and Design Councillor** Jan 2020 - Apr 2022
*Carleton University Student's Association Inc.* *Ottawa, ON*
- Represented the students of Engineering and Design by being the voice of their issues and services
- Engaged in council meetings and discussed solutions facing the student body

## Technical Skills

**Languages**: Python, Java, C, C++, C#, Javascript, SQL, VBA, HTML/CSS
**Frameworks**: Node.js, React, Django, Spring, Flutter, Bootstrap, Stripe
**Developer Tools**: Git/Github, Jira, Docker, Kubernetes, AWS, Azure, LaTeX, VS Code, Emacs, IntelliJ, QNX RTOS, Raspberry Pi, Microsoft Access, Power BI